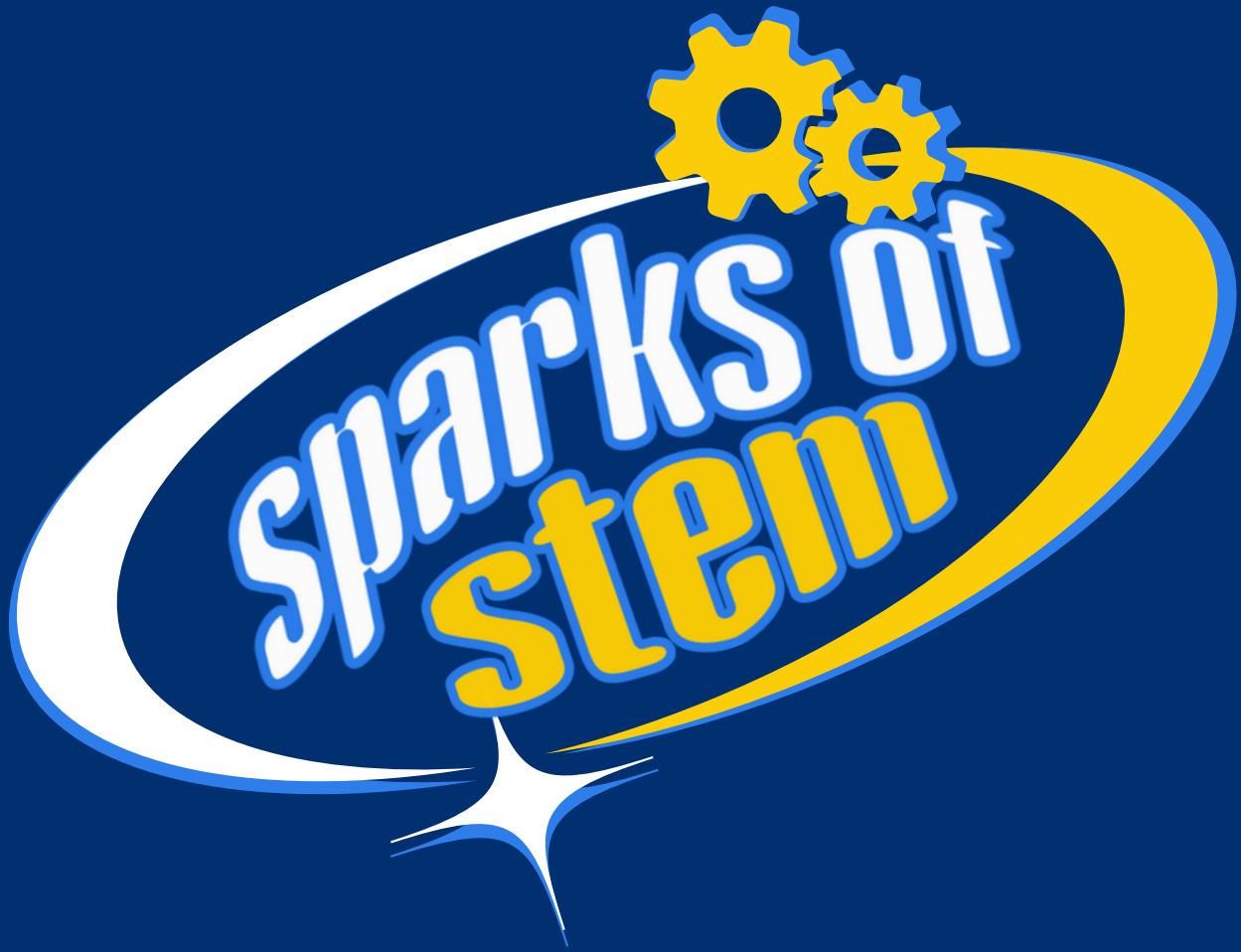




Community Awareness Project

**Sunset High School
13840 NW Cornell Rd
Portland, OR, 97229
Helen Wang & Angie Wang
12 January 2024**

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I. EXECUTIVE SUMMARY

PROBLEM

Many students face disparities in access and participation in STEM and CS courses, limiting their ability to learn.

SOLUTION

The Sparks of STEM (SoS) initiative: bringing interactive STEM learning sessions to classrooms, teaching basic skills, fostering collaboration and critical thinking, and inspiring interest and participation in STEM and CS.

MISSION

Sparks of STEM aims to inspire students to participate in STEM and CS courses one session at a time.



Sparker team Overcharged at the Code Dance Create Camp.



TARGET AUDIENCE

- **Primary:** Minority students who lack representation in STEM courses.
- **Secondary:** Rural and small schools where students haven't been exposed to STEM before

PROJECT BUDGET:

\$5.5k

raised through sponsorships and fundraising.

PROJECT GOALS

1

Provide sessions to 400 students

2

Reach underserved communities

3

Share the importance of STEM and CS education

PROJECT MILESTONES

Milestone 1

Host pilot sessions within the Beaverton School District to spark interest in STEM, gauge interest, and adjust lesson plan.

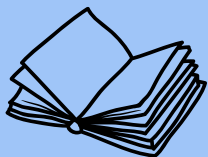
Milestone 2

With the Portland Metro STEM Hub, bring SoS to all schools within the Portland Metro Area (an addition of 4 school districts).

Milestone 3

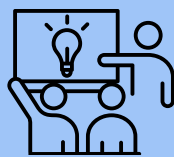
Reach out to the 13 other Regional STEM Hubs and establish "hubs" to bring SOS to schools all across Oregon.

PROJECT ACTIVITIES



Development of curriculum

Collaborate with educational professionals to create a unique hands-on approach to promote collaboration and critical thinking.



Hosting sessions

1. Schools (classes, afterschool clubs, STEM fairs)
2. Joint Sessions with Collaborators (STEM Like a Girl, ChickTech, Code and Dance Camp)
3. Community events (county fairs, markets)



Establishing other Sparks of STEM “Hubs”

Connect with regional STEM hubs and students to broaden our reach, encourage more Sparkers to join our cause, and ensure continuance.

QUALITY MANAGEMENT PLAN

1. **Check-up meetings:** Weekly meetings with “Sparkers”, collaborators, and the Portland Metro STEM Hub.
2. **Surveys/Feedback:** We use end-of-session feedback forms to improve our sessions.
3. **“Sparkers” Training:** All “Sparkers” are trained by us to prepare for sessions.
4. **Quality Management tools:** Notion and Google Calendar are used for efficient communication.

RECCOMENDATIONS FOR FUTURE PROJECTS

1. **Long-term engagement project:** Establish a mentorship program.
2. **Technology/Engineering Career Fair:** Organize career fairs inviting industry professionals and robotics teams.
3. **Online STEM Tutorials:** Provide tutorials ooprogramming and utilize CAD (computer-aided design)!

KEY METRICS

20

Schools Visited

600+

Students Impacted

8

Forged Collaborations

\$5.5K

Acquired and Fundraised

80%

of Students were More Interested

II. INITIATING

A. Statement of Problem

With the rise of new smart technologies like artificial intelligence, the importance of possessing knowledge in technology and engineering becomes more vital than ever. Yet across the nation and especially in Oregon, there is still a disparity in access and participation in these courses, specifically computer science (CS).

According to the Oregon Department of Education, many students are not provided with technology and engineering courses. For instance, “only 60% of public high schools in Oregon offered a foundational computer science course”, with the majority of these courses offered in urban areas compared to rural communities (Oregon Department of Education 2023).

Furthermore, underrepresented communities such as female-identifying students, students of color, students with disabilities, multilingual students, and students attending rural or small schools are much less likely to take STEM courses even when offered due to the intimidation of lack of representation, with only “**1 Black/African American** student in Oregon [taking] the AP Computer Science exam”, and only “**1 in 4 students** in a computer science class [being] **female-identifying**” (Oregon Department of Education, 2023).

These statistics continue to stem beyond the classroom, and into higher education and the industry.

Impacts:

- A continuous cycle of lack of representation in the STEM industry
- Students won’t possess crucial knowledge, stunting them in later education
- Lack of confidence in students that reflects in other aspects of their lives
- They won’t have access to many opportunities due to their lacked skill-set



6% of high school students took foundational computer science in 2021-22 in Oregon (CODE Advocacy Coalition)

B. Project Scope and Expected Benefits

When volunteering at local schools, we discovered a lack of access and participation in STEM classes among students. After talking to them, we found out why. Those without access were unaware they could take these courses and struggled to find alternative learning opportunities. Meanwhile, students with access found STEM classes intimidating due to false assumptions and lack of demographic representation. Existing solutions like company visits lack the excitement and reliability that students need, and no solutions currently address access and participation issues simultaneously.

Hence, as students ourselves, we believe **we can offer what others cannot: the passion and experiences of a fellow student**. Because we wholeheartedly believe in the significance of STEM education and can personally attest to how crucial it has been in our own experiences, we can successfully convey this message on a more personal level to students, and become the representation they seek.

The purpose of **Sparks of STEM (SoS)** is to teach STEM and computer science principles to students in a **fun and interactive way**, inspiring students to participate in these courses or programs outside of school. We aim to accomplish this by hosting interactive sessions at schools all across Oregon, providing personal experiences, interactive activities, and resources for them to use in the long run. Through these sessions, the expected benefits are that students will gain basic technical skills and the confidence to take STEM courses, recognize the fun in STEM, and utilize what they’ve learned to **empower themselves**, both in school and in the future.

III. PLANNING AND ORGANIZING

A. Project Goals

Our project has **3 main goals**, and we use them as benchmarks to ensure we stay on track to impact a diverse group of students and inspire them to engage in STEM.

1 PROVIDE SESSIONS TO 400 STUDENTS

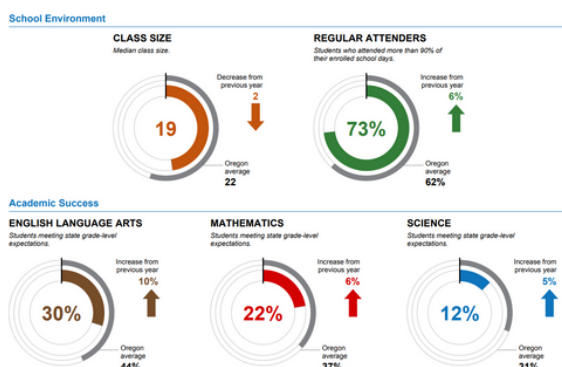
Visiting over 400 students ensures that we will reach a variety of schools, and interact with students from different demographics. A study by the University of Nevada, Reno emphasized the **importance of collaborative hands-on activities** as an important strategy to support a student in STEM (Wiest, L. R. 2014). Implementing that into our sessions, we brought a variety of activities, tailored to the needs and interests of students. For younger students, we taught basic lever systems through DIY candy catapults, and for older kids, we taught programming with basic loops, and they coded their LED vests (see appendix for a full list of activities).

Additionally, we reached out to local organizations like STEM Like a Girl, Chick Tech, and Engaging Every Student to work on bringing STEM education to underserved communities. They were all very excited to join in our cause, and we hosted joint sessions with them to reach a wider audience.

2 REACH UNDERSERVED COMMUNITIES

To ensure we visited places with the least exposure to STEM education, the PDX STEM Hub offered us advice and resources to determine these “target” schools. We used the **At-A-Glance School (left) software** that provides the percentage of academic success, demographics, and those who qualify for free/reduced lunch. We also used the **ODE Enrollment Sheet** with detailed demographic statistics that allowed us to visit schools with a high Hispanic/Black population, special education, and English 2nd-language classes.

We also collaborated with organizations like STEM Like a Girl which provides opportunities for girls, and Engaging Every Student which provides opportunities for students from low-income communities.



At-A-Glance profile for Rigler Elementary

3 SHARE THE IMPORTANCE OF STEM AND CS EDUCATION

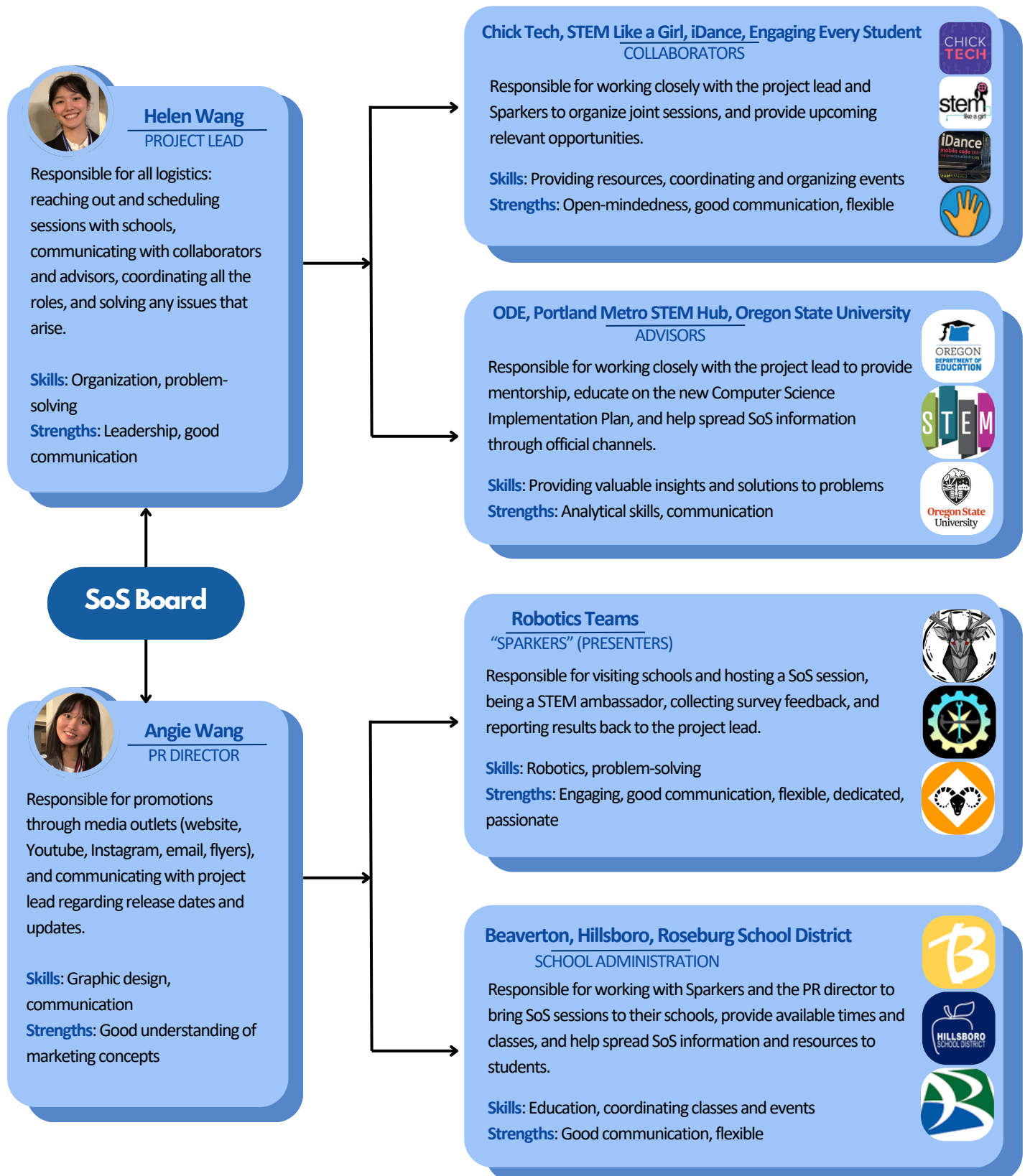
We aim to not only teach students and demonstrate how exciting STEM can be but also educate them about its importance in **fostering important skills** that can be used in both schools and finding future jobs. Sparkers shared personal stories on how STEM has positively impacted their lives, and informational videos were shown.

To ensure that students **continue to learn beyond our direct involvement**, we provided resources for students to explore so they could continue their STEM adventure. For example, we provided links to free software to learn programming, and tutorials to DIY projects to develop creativity and engineering skills. Lastly, we connected them with organizations that share local workshops and company tours for them to attend.



During a joint session with the Code, Dance, Create camp, students collaborated to design the best candy catapult, building teamwork and design skills.

B. Human Resource Management Plan



C. Schedule

i. Milestones (M.S.)

Milestone 1

Host pilot sessions within the Beaverton School District to spark interest in STEM, gauge interest and adjust lesson plan.

Milestone 2

With the Portland Metro STEM Hub, bring SOS to all schools within the Portland Metro Area (a total of 4 school districts).

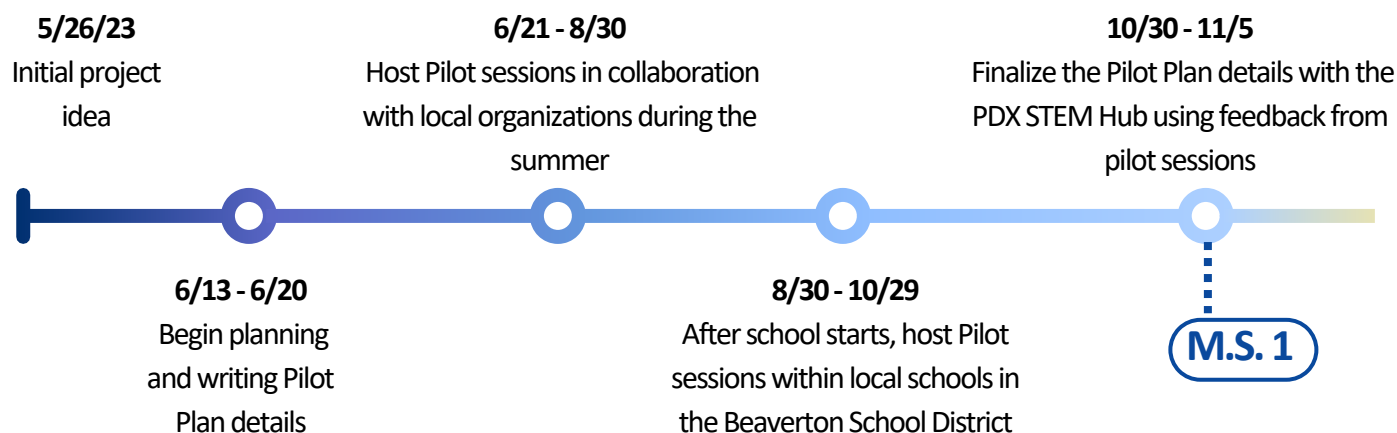
Milestone 3

Reach out to the 13 other Regional STEM Hubs and establish “hubs” to bring SOS to schools all across Oregon.

ii. Timeline

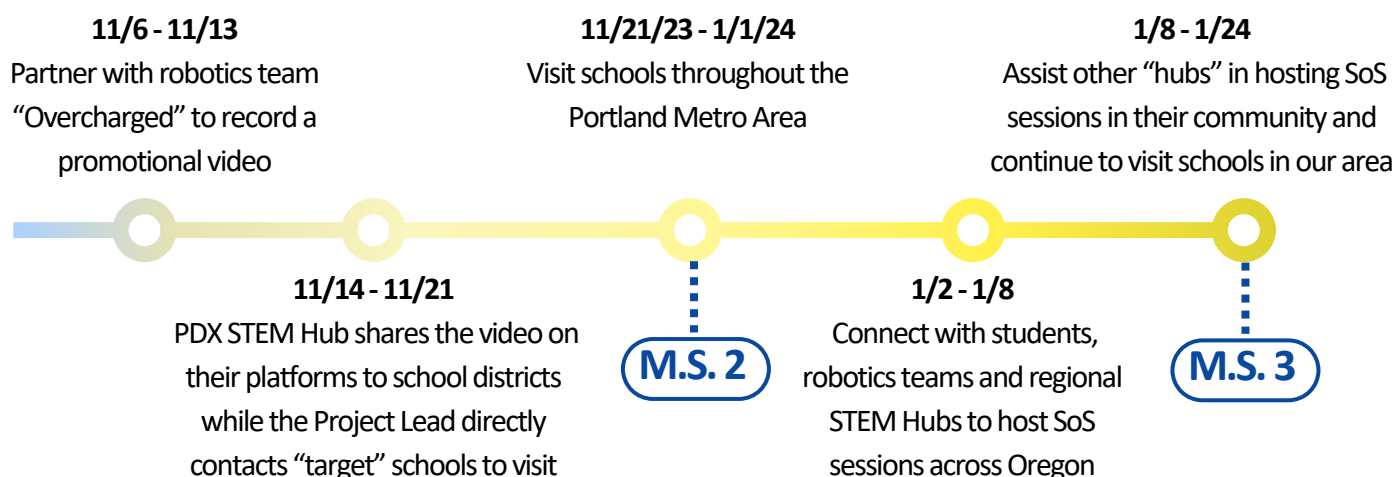
STAGE 1 *Planning*

STAGE 2 *Pilot Launch*



STAGE 3 *Expansion (Portland Metro)*

STAGE 4 *Expansion (Oregon)*



D. Quality Management Plan

Check-up Meetings

- We have **weekly meetings** with sparkers to receive updates on their sessions.
- We have weekly meetings with the Portland Metro STEM Hub to recap our progress, the budget state, and future steps.
- We meet with collaborators for updates on potential joint sessions.

Surveys/Feedback

We created **surveys for both the teachers and students** to fill out after a session. From the teacher's reply, we can determine our customer support and communication quality, and from the student's reply, we can gauge how engaging our session was, and how effective it was at increasing their interest in STEM.

"Sparker" Training

To ensure that Sparkers are able to host quality sessions on their own, we go through **online training**, teaching them about the curriculum, the goal, and some basic problem-solving skills in case something happens (eg. someone gets hurt, or something stops working).

Project Management Tools

We utilize many software applications to make **communication quick and consistent**. For example, Notion is used to keep track of tasks and brainstorming ideas; Google Calendar is used to manage meeting and session times; and Instagram's analytics report is used to measure and retain engagement on our platform.



In a meeting with robotics teams Aires and Cybernetic Elks from Roseburg, we discussed the procedure for hosting a session and the possibility of establishing a Sparks of STEM hub in their community.

Additionally, team Aires provided us with their "FUSE" circuit kits, which we use during sessions to teach basic electrical engineering to classes.

E. Risk Management Plan

When we visit classrooms, we interact with young students so there are increased risks. Therefore we have a **set protocol** to follow, ensuring both the safety of “Sparkers” and students. We have implemented the following response strategies to potential issues that might arise.

Description of Potential Issue	Potential Impact	Response Strategy	Owner
Students getting injured during a session	Liability consequences and damage of reputation	Sparkers will utilize the first-aid kits they carry to treat any minor injuries that arise, or locate the nearest nurse’s office for more severe injuries.	Sparkers
Sparkers unable to make it to a scheduled session	Last minute adjustments are needed (ex. distribution of tasks), and possible rescheduling if more than one Sparker is absent	All Sparkers undergo cross-training to be familiar with all aspects of the session, and are capable of replacing and leading whenever needed. Hence, the remaining Sparkers simply carry on with the session, and fill in the empty positions.	Project Lead
Shortage of knowledgeable, skilled Sparkers	Constraint of number of schools we are able to visit at a time, or the quality of a session is diminished	Above all, we maintain the idea of quality over quantity, and providing the best experience for the students, rather than rushing to reach a larger audience. If this problem occurs, we will shift our focus to host larger sessions (such as school assemblies or STEM nights), so a single team can have a greater impact.	Project Lead
Weather	Delay of schedule	Clear communication and an organized schedule is previously established between schools and Sparkers, so the Sparker simply reschedules to the nearest available time slot.	Project Lead
Lack of interest from students	The goal of the session is not reached, and students remain disinterested in STEM	After the session, we will adjust our curriculum based on feedback, and improve our tailorability for future schools.	Sparkers & Project Lead

F. Proposed Project Budget

After the pilot plan was released, we contacted and acquired many **potential investors** by pitching them our initiative and the predicted impact. We were able to gain the majority of our capital from the investors, while other material and labor costs were covered by forged collaborations with local organizations. These **partnerships** provided us with a variety of activities we could provide during our program, such as circuit kits, LED vests, and more. Additionally, all of the Sparkers are volunteers, from robotics teams, and local organizations like STEM Like a Girl and ChickTech.

i. Projected Expenses

Expenses	Unit	# of Units per session*	Unit Rate	Cost
Activity 1: Robot Demonstration				
1.1 Robotics team "Sparkers"	Per person	3	In-kind cost	\$0
1.2 Transportation	Per person	3	In-kind cost	\$0
1.3 Meals	Per person	3	In-kind cost	\$0
Total Activity 1				\$0
Activity 2: Spaghetti/Marshmallow Tower				
2.1 Dry Spaghetti	Per item	4	\$1.50	\$6.00
2.2 Marshmallows	Per item	4	\$1.59	\$6.36
Total Activity 2				\$12.36
Activity 3: Candy Catapult Challenge				
3.1 Popsicle Sticks (Pack of 1000)	Per item	1	\$6.39	\$6.39
3.2 Plastic Spoons (Pack of 100)	Per item	1	\$1.99	\$1.99
3.3 Rubber Bands (Pack of 100)	Per item	1	\$2.99	\$2.99
3.4 Candy (Pack of 100)	Per item	1	\$9.48	\$9.48
Total Activity 3				\$20.85
Activity 4: Cardboard Robot Hand				
4.1 Cardboard Tubes (Pack of 12)	Per item	2	\$9.99	\$19.98
4.2 Drinking Straws (Pack of 200)	Per item	1	\$6.55	\$6.55
4.3 Roll of String	Per item	1	\$1.99	\$1.99
4.4 Needle (Pack of 10)	Per item	1	\$3.49	\$3.49
4.5 Paper Clips (Pack of 1000)	Per item	1	\$1.57	\$1.57
Total Activity 4				\$33.58
Activity 5: Special Activities				
5.1 FUSE Circuit Kit (Pack of 10)	Per item	2	In-kind cost	\$0
5.2 LED Vests Kit (LED Vest + Ipad)	Per item	24	TBD	\$0
Total Activity 5				\$0

*assuming a class has an average of 24 students

ii. Income

Source	Status	Amount
Engaging Every Student	Acquired	\$250
Ignited Tech Non-Profit	Acquired	\$750
Portland Metro STEM Hub	Acquired	\$100
TOTAL PROJECT INCOME:		\$1,100

The average cost to host an activity is **\$22.26**. Currently, we have hosted 20 sessions for a total of **\$445.27**, which is around **40.48%** of our budget. This amount will vary depending on which activities we end up offering.

Our **highest-value activity** is the Spaghetti/Marshmallow Tower because one session costs \$12.36, which can be further reduced by splitting the students into groups. But due to the amount of effort it takes to clean up, we predicted it wouldn't be a popular activity that teachers would select.

Our **least-value activity** is the Cardboard Robot Hand, because a single set costs \$33.58, and this is an individual activity. Due to its complexity, we predict that it will be mainly older students who would be interested in this activity.

We predict that our most **popular activity** will be the Candy Catapult Challenge, as we received the most positive feedback for it from our pilot session. Additionally, it's not too complex nor messy to clean up, and perfect for all age groups.

We are working to establish and retain strong relationships with our current sponsors. This way, for future large projects, we can have a stable sponsor to fund our initiatives.

IV. EXECUTION

A. Implementation

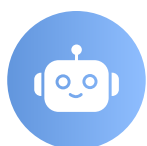
Development of Curriculum

When planning our session plan, we realized the most important aspect was our curriculum, and how we would **differentiate** it from a normal class environment. Unlike a traditional classroom where a teacher lectures for hours, we emphasized **hands-on learning** through interactive activities, which allowed for increased student concentration, retention of information, and interest in pursuing STEM. Additionally, we **tailored our curriculum** for different grade levels, what they were learning, and the number of students.

This allowed us to formulate the best teaching lesson for all our sessions, with the following components:



- Short 15-20 min **presentation** sharing our personal experiences in STEM and fun videos
- Emphasizing the importance of STEM education
- Q&A session



- A **robot demo** from Sparkers' robotics teams with a walkthrough of the mechanics behind each robot component
- A chance for students to drive it!



- A variety of **STEM activities** to encourage creative experiences aimed at providing a collaborative, hands-on activity for students to practice their STEM and critical thinking skills
- Our different activities vary in difficulty, and technical area, allowing for more flexibility and adaptability for schools



- Many students don't have STEM courses offered in school, so we provided a flyer with **resources** they can explore out of school
- We included websites to learn basic programming and design, STEM events hosted by local organizations, visits to companies, and other community events hosted by robotics teams



- We collaborated with the ODE, helping them by advocating for the **Computer Science Implementation Plan**, which plans to implement CS classes in schools all across Oregon by 2027-28
- Not only do we share about this exciting event, but we also encourage students to take them when offered and give them the confidence to tackle the courses



Sparkers demo-ing their robot at Findley Elementary School's STEM night.

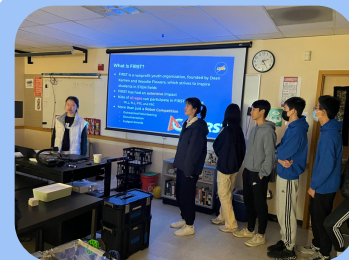
Establishing other Sparks of STEM "Hubs"

To ensure project sustainability and to broaden our reach, we decided to form Sparks of STEM "Hubs" across Oregon. We selected a team of Sparkers from different areas of Oregon who were already experienced in hosting their own sessions and helped them **establish their own system**. Through this, they would still report to us, but they could begin inviting other Sparkers within their area, and be in charge of visiting their local schools. This helped reduce our side of the workload, and at the same time, invite many other Sparkers to join in on the movement.

Hosting Sessions

After the Portland Metro STEM Hub sent out the informational video on our behalf to the entire Portland Metro School District, we immediately garnered interest and began organizing visits to schools. There were 2 methods of signing up for a session: emailing us or registering through our website (see appendix). Just only 3 days after the email was sent out, we already had 8 schools interested in having us visit. Currently, we offer two types of sessions:

a. **School Lessons** - We visited classrooms, after-school clubs, STEM nights, and more. Usually, the teacher (who taught STEM or STEAM) would allocate a portion of their class period for our interactive session, and we would stay for multiple classes.



A session hosted at the International School of Beaverton's engineering club.



Students wore the LED vests they programmed and danced together!

b. **Joint Sessions with Collaborations** - This mainly took place during the summer when organizations hosted STEM camps. Many of the camps had a section reserved for guest speakers sharing about different industries, in which case we'd get invited, and Sparkers would share about their robotics program and plans for the future. Since the event was collaborative, we would stay behind and assist in helping the students with their projects, and connect with the local community.

B. Promotion

The promotion of SOS was done mainly through digital media, as that was the best method to communicate with schools all across Oregon. Additionally, the majority of the initial advertising was done through the local robotics team, Overcharged's platforms, as they already had an established platform with the relevant audience. The following platforms were used:



Website: The established website for the initiative allowed an easy read for people to learn more about the cause, and sign up to show interest.



Instagram: It is used to post updates on schools visited, share fun facts about STEM and CS education in Oregon, and is also a method for people to learn more about the initiative. We use the "Professional dashboard insights" to observe and tailor our content to our target audience.



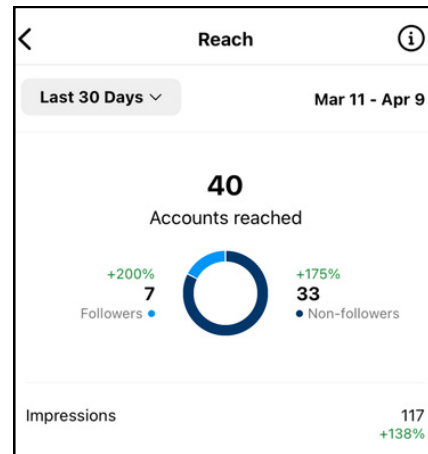
Email (view appendix): This was used by the Portland Metro STEM Hub to send out mass emails to schools, and was also the main method of communication between schools and Sparkers.



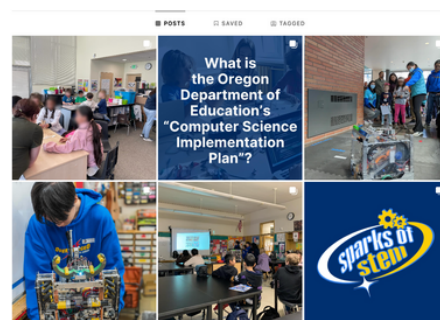
Flyers (view appendix): Physical flyers were passed out during community events to promote the initiative, encourage prospective Sparkers to sign up, and for potential schools who want to request a visit.



Word-of-mouth: After successful sessions, teachers often shared about us with their colleagues their colleagues, recommending our program.



Instagram analytics and posts



C. Expenses

Since all the Sparkers were volunteers, the budget was significantly reduced as transportation and meals were all self-covered. The main cost came from the materials needed to host the various activities we offered. To reduce costs, we bought materials in bulk beforehand, anticipating the demand we would receive. Hence, many of the activity materials can be reused for 2-3 sessions.

Overall, the **total expense came out to \$404.63**, which is **under 40% of our funds**. We are considering implementing new activities and removing the more expensive and less popular options to be more cost-effective.

Expenses	# of Units	Unit Rate	Cost
Personnel			
Robotics Team	20	In-kind cost	\$0
Materials			
Activity 1	18	In-kind cost	\$0
Activity 2	2	\$12.36	\$24.72
Activity 3	15	\$20.85	\$312.75
Activity 4	2	\$33.58	\$67.16
Activity 5	1	In-kind cost	\$0
Other Costs			
Participant Transportation	20	In-kind cost	\$0
Meals	20	In-kind cost	\$0
TOTAL PROJECT COST			\$404.63

V. MONITORING AND CONTROLLING

A. Monitoring

i. Schedules

We utilize a shared Google calendar system, displaying no-school days, meetings with potential collaborators, and scheduled sessions. By labeling the event with relevant information, we are all informed of whether a session is rescheduled, the location of the event, the overall schedule of that day (which classes visited), and more. Sparkers can access the calendar to remind themselves of upcoming events, and to aid in scheduling sessions with schools.

15	16	17	18	19
No school	No school	Snow day - No school	Snow day - No school	
Martin Luther King Jr. Day	9:15am [Rescheduled] Astor K-8			
2pm Collab - All Hands on Tech Robot				
3pm Collab - Aires Robotics				
22	23	24	25	26
Finals week (early release)				No school
		1pm Collab - Darryl LED Vests (WOU)	9:15am Astor K-8	9:15am Harrison Park

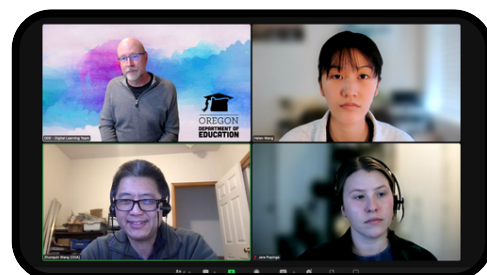
Our google calendar, listing no-school days, cancelled/rescheduled sessions, and meetings with collaborators.

ii. Budget

The project lead manages the budget which is then shared with advisors for suggestions. Basic materials needed for the sessions are bought separately by teams to reduce complications and are then sent in for reimbursement to the project lead. Previously acquired sponsorships cover the fees; currently, more than 80% of the initial funding is left. Future capital will be **acquired through business pitches** to local organizations and industry companies, and through outreaches.

iii. Project Quality

- **Quality of Sparkers:** To ensure the quality of the Sparkers, the project lead works closely with them, getting them informed of the Sparks of STEM initiative, and working through preliminary training, safety training, and the overall process from organizing a session to executing it. Therefore she can ensure the projected quality of the project.
- **Quality of session:** By providing teachers and students with a survey, we can collect relevant information to gauge the interest and impact of the session from the students, and the ease of the scheduling process with the teacher (see appendix).
- **Quality of overall project:** The project lead organizes an after-session meeting with teams to review how successful the outreach was, as well as the collected feedback from both the school and the Sparkers themselves. She also hosts weekly meetings with the Portland Metro STEM Hub to summarize the progress and adjust the plan as needed. Topics discussed include the state of budget (whether we're over/under budget), and allocation of workforce allocation. Through all of this, the hope is to ensure initial project quality and continue improving it.



Meeting with ODE where we recapped our progress and discussed the details and plans on advocating for the CS Implementation Plan.

B. Controlling

The primary obstacle we faced was time availability for Sparkers to visit schools to host SOS sessions. For maximum exposure, the best time to host a SOS was during school hours, but as students themselves, Sparkers couldn't go as they had school too. This problem was dealt with a preventive and an alternative solution. We offered afterschool sessions as an option (eg. fairs, STEM nights, after-school clubs) so we had more flexibility when organizing times. In addition, we **utilized no-school days** within our school district to visit other school districts while they were still in school. For example, with an average of 2-3 no-school days per month, the Portland Metro team was able to attend sessions while staying in school.

Another issue encountered was the overwhelming demand, especially during the initial release. Due to the restriction of time availability from Sparkers who were still in school, only 2-3 during-school SOS sessions were able to be hosted, decreasing the efficiency. To solve this issue, we recruited more teams within the same area. For example, the robotics team All Hands on Tech was invited to participate and host their own sessions.

Lastly, during the middle of January, there was a snow blizzard in the Portland area, and schools were shut down. Coincidentally, an SOS was scheduled for that week. Thanks to **quick and effective communication** with the school, the session was moved to the next week.



During the summer, we visited the Code, Dance, Create camp, teaching 40+ kids how to code through our session!

VI. CLOSING THE PROJECT

A. Evaluation of Key Metrics

20

Schools Visited

600+

Students Impacted

8

Forged
Collaborations

\$5.5K

Acquired and
Fundraised

80%

of Students were
More Interested

Connecting with 3 STEM Hubs, 6 school districts, and impacting a diverse community of students (girls, SPED, ELD, rural areas, low-income areas, minority races), we are proud to say that **we left a smile on 100% of the students**. Not only did we share about the importance of STEM, but we emphasized how fun it could be, and provided them with resources to continue learning even when we leave their school. Through our efforts, we hope to inspire the next generation of STEM!

“This is the most fun we’ve ever had in the classroom!”

- Students at Astor K-8 School

B. Lessons Learned

From this project, we learned the importance of **effective planning and flexibility**. We found success in using a shared calendar that allowed us to coordinate schedules and availability among volunteers and collaborators and reschedule sessions, which came in use when several sessions had to be adjusted to accommodate a sudden winter storm.

We had to make adjustments during sessions to **adapt to different students**, and the situation at hand. For example, if there was a younger audience, we had to shorten our presentation portion of the lesson to keep them engaged.

Lastly, we also had to reach out to a variety of local organizations to establish connections and make Sparks of STEM sessions possible. This taught us the **importance of networking and building relationships** within the community. Hence, we are looking to recruit a designated outreach lead to increase our impact!



We attended OMSI’s festival, teaching children how to drive the robots, and hosting fun activities.

C. Recommendations for Future Projects

Long-term engagement project: For students who show a growing interest in technology and engineering, we want to establish a mentorship program. We hope a longer-term program can help sustain the student’s enthusiasm and engagement in technology and engineering courses.

Technology/Engineering Career Fair: With the aid of local organizations, such as STEM Like a Girl and the Portland Metro STEM Hub, we can organize a career fair inviting industry professionals and robotics teams. This will allow students to gain insights into various career paths and understand the real-world applications of STEM.

Online STEM Tutorials: Many students expressed interest in watching tutorials on how to program and utilize CAD (computer-aided design). This allows us to broaden our reach worldwide and provide accessibility for students!

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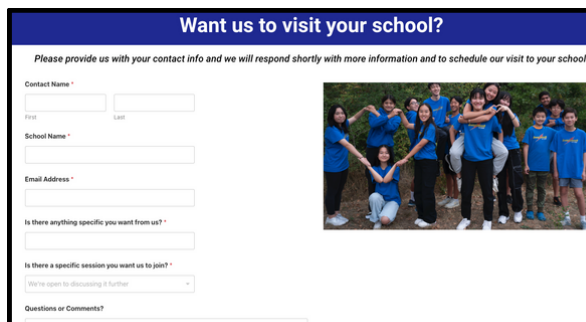
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Activities List

- Robot demonstration and game
- Candy Catapult Challenge
- Spaghetti/Marshmallow Tower
- Cardboard Robot Hand
- FUSE Circuit Kit
- LED Vests

Website

Registration form (<https://overchargedrobotics.org/sparks-of-stem/>)



Want us to visit your school?

Please provide us with your contact info and we will respond shortly with more information and to schedule our visit to your school!

Contact Name *
First Last

School Name *

Email Address *

Is there anything specific you want from us? *

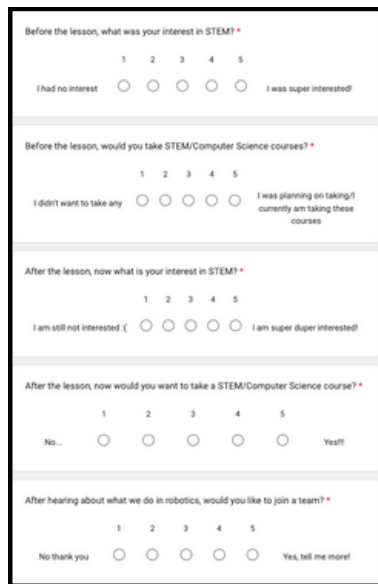
Is there a specific session you want us to join? *

We're open to discussing it further

Questions or Comments?

Survey for students

The survey for students consisted of 5 simple 1-5 scale questions, allowing us to easily determine how successful our session was. Results showed that **80% of students had an increased interest in STEM** after our visit.



Before the lesson, what was your interest in STEM? *

1 2 3 4 5

I had no interest I was super interested!

Before the lesson, would you take STEM/Computer Science courses? *

1 2 3 4 5

I didn't want to take any I was planning on taking/I currently am taking these courses

After the lesson, now what is your interest in STEM? *

1 2 3 4 5

I am still not interested I am super duper interested!

After the lesson, now would you want to take a STEM/Computer Science course? *

1 2 3 4 5

No... Yes!!

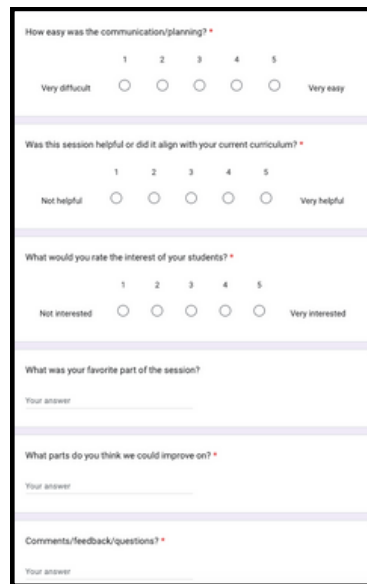
After hearing about what we do in robotics, would you like to join a team? *

1 2 3 4 5

No thank you Yes, tell me more!

Survey for teacher

The survey for teachers is a longer and more detailed survey, covering feedback from planning, alignment with curriculum, etc. We used this feedback to constantly improve our scheduling and communication methods.



How easy was the communication/planning? *

1 2 3 4 5

Very difficult Very easy

Was this session helpful or did it align with your current curriculum? *

1 2 3 4 5

Not helpful Very helpful

What would you rate the interest of your students? *

1 2 3 4 5

Not interested Very interested

What was your favorite part of the session?

Your answer

What parts do you think we could improve on? *

Your answer

Comments/feedback/questions? *

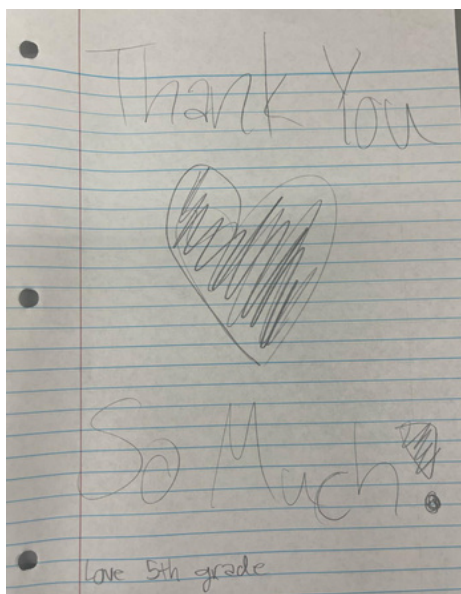
Your answer



Promotional Video

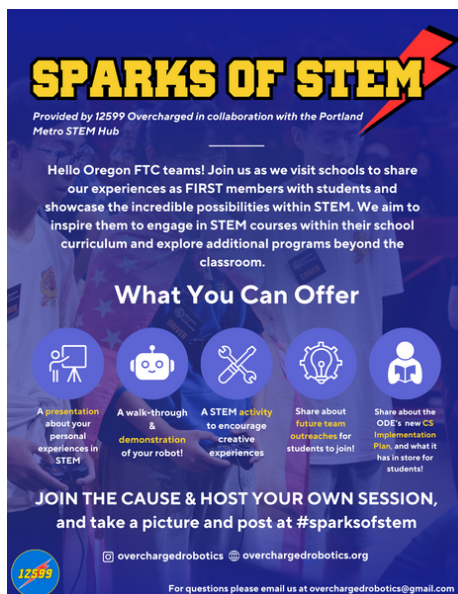
We partnered with a local robotics team “Overcharged” to post a video sharing about the Sparks of STEM initiative and shared it through the Portland Metro STEM Hub to schools across our STEM district. Later on, other regional STEM Hubs continued to spread our videos to their districts and schools.

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Thank You Note

"Thank you" note from Rigler Elementary students!



Flyer for "Sparkers"

This is an example of a flyer shared with prospective Sparkers detailing how to host their own sessions.



Session Plan for Sparkers

To make the process of hosting a Sparks of STEM session easier, we compiled all the components needed to host a successful session and shared this document with Sparkers as a part of the introductory training.

Hello! My name is Helen Wang, a junior at Sunset High School, and captain of 12599 Overcharged, a globally recognized robotics team from Portland, Oregon! As students who have benefited from STEM education, we deeply understand the significance of early exposure to these courses and programs. However, we are aware of the barriers students face, such as limited access and a lack of confidence when it comes to choosing STEM and CS courses. That's why I am excited to share our initiative, "Sparks of STEM", aimed at addressing this issue!

In collaboration with the Portland Metro STEM Hub, we are launching "Sparks of STEM" to reach schools throughout Oregon! This initiative will provide interactive learning sessions where we'll not only share our experiences but also showcase our world-ranking robot. Additionally, we'll present a captivating engineering challenge, and provide resources for students to explore on their own. If you are interested in learning more, please watch this video, visit our website, or contact us at overchargedrobotics@gmail.com. We look forward to visiting your school to spark more engagement in STEM among your students!

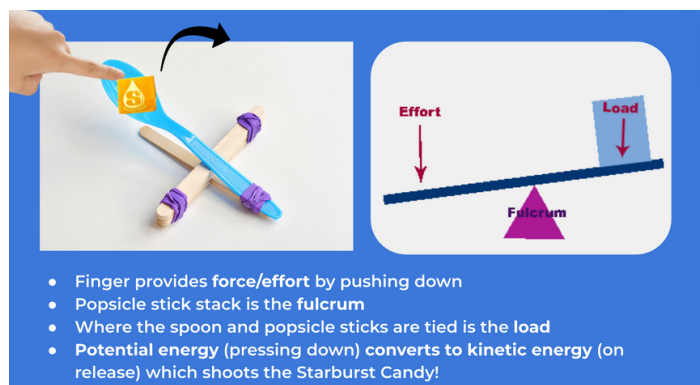
Best Regards,

Dave
David Strayer
503-860-6055
PMSP "Computer Science for All" Project Coordinator
pdxstem.org/computer-science



Sample Email

This is an example of an email sent out by the Portland Metro STEM Hub on behalf of us to spread Sparks of STEM sessions to schools all across Oregon.



- Finger provides **force/effort** by pushing down
- Popsicle stick stack is the **fulcrum**
- Where the spoon and popsicle sticks are tied is the **load**
- **Potential energy** (pressing down) converts to **kinetic energy** (on release) which shoots the Starburst Candy!

Lesson Plan

Along with our fun activities, we included lesson plans that taught the concepts behind how they worked. For example, for our candy catapult, we used that to explain how levers worked, giving them exposure to mechanical engineering.